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Decomposition AI for thermal satellite images resolution enhancement

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Within:



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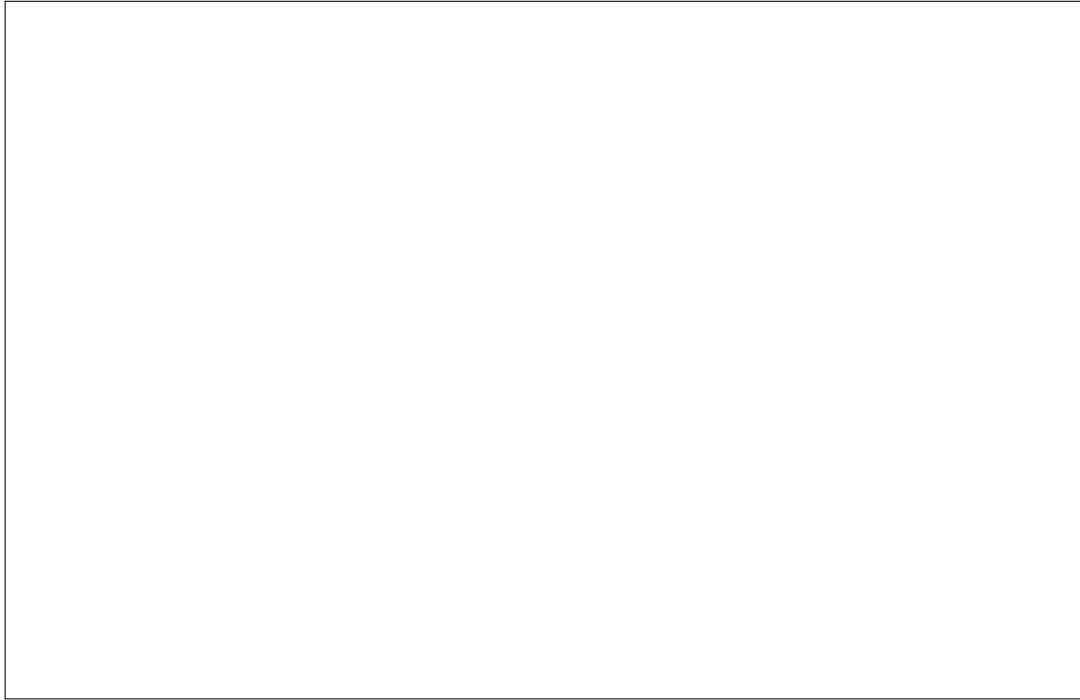
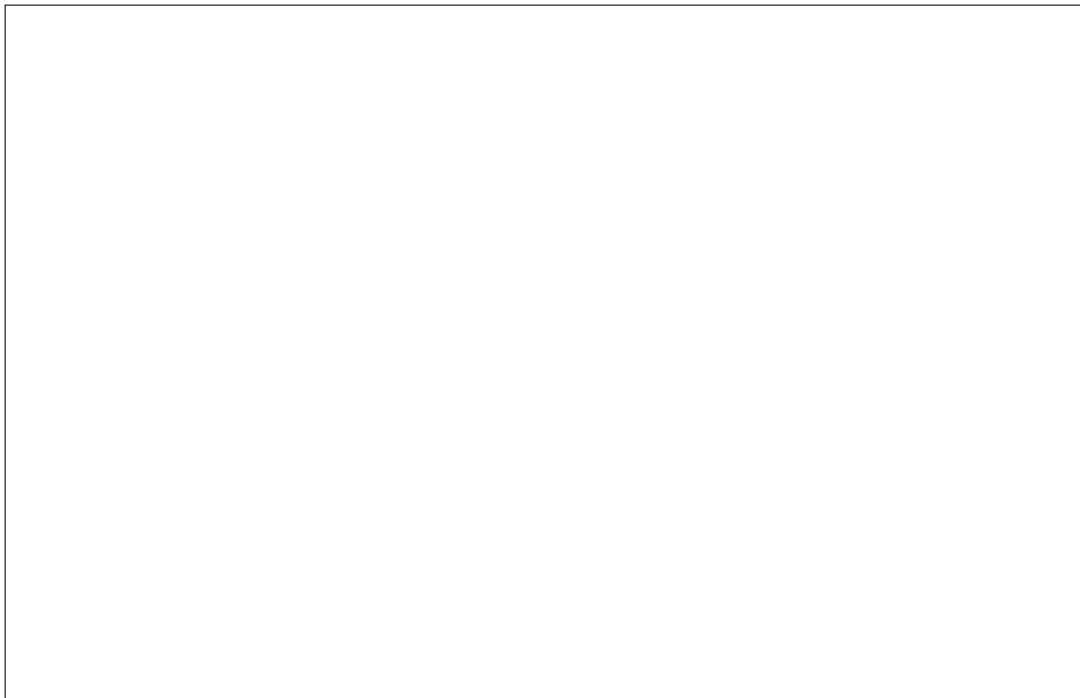
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Abstract

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List of Abbreviations

AI	Artificial Intelligence
AI4DM	AI for Decision Making Lab, Concordia University
API	Application Programming Interface
BM25	Best Match 25, a probabilistic lexical ranking function
ColBERT	Contextualized Late Interaction over BERT
DAG	Directed Acyclic Graph
DPR	Dense Passage Retrieval
EPT	Tunisia Polytechnic School (<i>École Polytechnique de Tunisie</i>)
EV	Engineering, Computer Science and Visual Arts Complex, Concordia University
FDR	False Discovery Rate
FLARE	Forward-Looking Active Retrieval Augmented Generation
IRCoT	Interleaving Retrieval with Chain-of-Thought Reasoning
LLM	Large Language Model
MDR	Multi-Hop Dense Retrieval
MIAE	Mechanical, Industrial and Aerospace Engineering, Concordia University
NLI	Natural Language Inference
PAC	Probably Approximately Correct
PPI	Prediction-Powered Inference
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RAG	Retrieval-Augmented Generation
TF-IDF	Term Frequency-Inverse Document Frequency
UQ	Uncertainty Quantification

General Introduction

Chapter 1

General Presentation and Context of the Internship

Introduction

The purpose of this chapter is to situate the project within its general research and institutional context, and to present the main objectives, research directions, and methodology that guided the work carried out during the internship.

The work reported in this thesis was conducted as a research internship at the AI for Decision Making Lab of Concordia University, under the supervision of Dr. Yassine Yaakoubi, and constitutes the final-year project of the engineering curriculum at École Polytechnique de Tunisie (EPT).

This chapter first introduces the host institution and the research laboratory in which the internship was carried out, together with their main areas of research. It then presents the broader scientific context of the internship, centered on large language models, retrieval-augmented generation, and multi-hop question answering. The progression of the research work is subsequently outlined, from the initial literature exploration and systematic study of trust signals in RAG systems, to the investigation of compositional question difficulty in language model representations, and finally to the development of a retrieval-oriented framework for multi-hop question decomposition. The chapter concludes by presenting the main objectives, contributions, and research methodology adopted throughout the internship.

1.1 Host organisation

The internship was hosted by the **AI for Decision Making Lab (AI4DM)**, a research group within the Department of Mechanical, Industrial and Aerospace Engineering (MIAE) of the Gina Cody School of Engineering and Computer Science at Concordia University, Montreal, Canada. Table 1.1 summarises the host organisation. The following subsections present the university, the faculty and department to which the laboratory belongs, and finally the research group in which the work was carried out.

Table 1.1: Host organisation details

Institution	Concordia University
Faculty	Gina Cody School of Engineering and Computer Science
Department	Mechanical, Industrial and Aerospace Engineering (MIAE)
Laboratory	AI for Decision Making Lab (AI4DM)
Supervisor	Dr. Yassine Yaakoubi
Location	Montreal, Quebec, Canada
Research axes	Optimisation, learning and simulation for large-scale decision making under uncertainty; large language models and retrieval-augmented generation; uncertainty quantification and reliability of AI systems

1.1.1 Concordia University

Concordia University is a public comprehensive research university located in Montreal, Quebec. It was officially founded in 1974 following the merger of Loyola College, established in 1896, and Sir George Williams University, whose origins can be traced to the educational activities of the Montreal YMCA [N1, N2]. The name of the university derives from the motto of the City of Montreal, *Concordia salus*, meaning “well-being through harmony.”

Concordia operates two main campuses: the Sir George Williams Campus in downtown Montreal and the Loyola Campus in the west of the city. In the 2024–2025 academic year, 43,922 students were enrolled in credit courses across the university [N3]. Its academic activities cover a broad range of disciplines through the Faculty of Arts and Science, the Gina Cody School of Engineering and Computer Science, the John Molson School of Business, and the Faculty of Fine Arts.

1.1.2 Gina Cody School of Engineering and Computer Science

The Gina Cody School of Engineering and Computer Science is Concordia University’s faculty dedicated to engineering and computer science. In 2018, following a fifteen-million-dollar donation from Concordia alumna Gina Parvaneh Cody, the faculty was renamed in her honour, becoming the first engineering faculty in Canada to be named after a woman [N4]. Gina Cody, who earned her doctorate in building engineering from Concordia in 1989 and was the first woman in Canada to obtain a PhD in the discipline, has served as Chancellor of Concordia University since January 2025 [N5].

The school brings together 11,315 undergraduate and graduate students, 262 faculty members, 43 undergraduate and graduate programmes, and 20 interdisciplinary research centres [N6]. Its academic structure comprises seven departments, including the Department of Mechanical, Industrial and Aerospace Engineering (MIAE), to which the host research group belongs.

Teaching and research activities are conducted principally in the Engineering, Computer Science and Visual Arts Integrated Complex (EV Building), located on Concordia’s Sir George Williams Campus and shown in Figure 1.1.



Figure 1.1: The EV Complex, home of the Gina Cody School (photograph: Tasitch, CC BY 2.0)

1.1.3 Research group, supervision, and research axes

The work presented in this report was conducted within the AI for Decision Making Lab (AI4DM), led by Dr. Yassine Yaakoubi, Assistant Professor in the Department of Mechanical, Industrial and Aerospace Engineering at Concordia University.

The research activities of the group lie at the intersection of optimisation, machine learning, and simulation for large-scale real-world decision making. The group combines mathematical optimisation methods, including mixed-integer and stochastic programming and decomposition techniques, with machine-learning approaches such as deep learning, reinforcement learning, and graph-based models. The general objective is to design decision-support systems that remain efficient, robust, and scalable in complex and uncertain environments, with applications including logistics, transportation, supply chains, and sustainability.

A major part of the group's work is organised around machine-learning-augmented optimisation, where learned models are integrated into classical optimisation procedures, and end-to-end optimisation learning, where learning mechanisms are used more directly to guide decision making.

A further strand of the group's research activity extends these concerns with robustness, uncertainty, and adaptive decision making to systems built on large language models. This includes retrieval-augmented generation, where language models are grounded in evidence retrieved from external knowledge sources, as well as uncertainty quantification, reliability assessment, and adaptive behaviour in such systems.

The work carried out during this internship belongs primarily to this latter research direction. It focuses on multi-hop question answering in retrieval-augmented systems and investigates, from complementary perspectives, how the difficulty of such questions can be understood and how their processing can be made more effective. The research therefore evolved from the study of reliability and actionable signals in RAG systems, to the analysis of compositional question difficulty in the internal representations of large

language models, and finally to the development of a retrieval-oriented approach to multi-hop question decomposition.

This research environment provided the basis for the successive studies conducted during the internship and for the formulation of the main research problem addressed in this thesis.

1.2 Problem statement

1.3 Objectives and expected deliverables

1.4 Working methodology

Conclusion

Chapter 2

Background and State of the Art

Introduction

Multi-hop question answering requires a system to retrieve and combine several pieces of evidence before producing an answer. In Retrieval-Augmented Generation (RAG), this creates a particular difficulty: the evidence needed for a complex question may not be easily retrieved from the original query, and different questions may require very different amounts of retrieval and reasoning.

This chapter introduces the concepts needed for the remainder of the thesis. It first presents RAG and multi-hop question answering, then reviews retrieval methods and the notions of evidence coverage and retrieval depth. It subsequently discusses adaptive computation and question decomposition, before focusing on decomposition as a means of making evidence easier to retrieve. The chapter ends by identifying the two research gaps addressed in the following chapters.

2.1 Multi-Hop Question Answering and Retrieval-Augmented Generation

2.1.1 Retrieval-Augmented Generation

Large Language Models (LLMs) acquire substantial knowledge during pre-training, but their parametric knowledge is neither a complete nor a continuously updated record of the world. It may omit specialised facts, become outdated, or be insufficient when a question requires information from a particular source. RAG complements the model with an external corpus that can be searched at inference time [1].

Let the corpus be

$$\mathcal{C} = \{d_1, d_2, \dots, d_N\}, \quad (2.1)$$

where each d_i is a document or passage. For a query q , a retriever assigns a relevance score $s(q, d)$ and returns

$$D_k(q) = \underset{d \in \mathcal{C}}{\text{TopK}} s(q, d). \quad (2.2)$$

The generator then predicts an answer y conditional on the question and the retrieved passages,

$$p(y \mid q, D_k(q)). \quad (2.3)$$

This division of labour is central: the retriever determines which external information becomes visible, while the generator interprets and combines that information. Figure 2.1

depicts the separation between the two components. A stronger generator can reason more effectively over retrieved context, but it cannot directly exploit a relevant passage that the retriever never exposes.

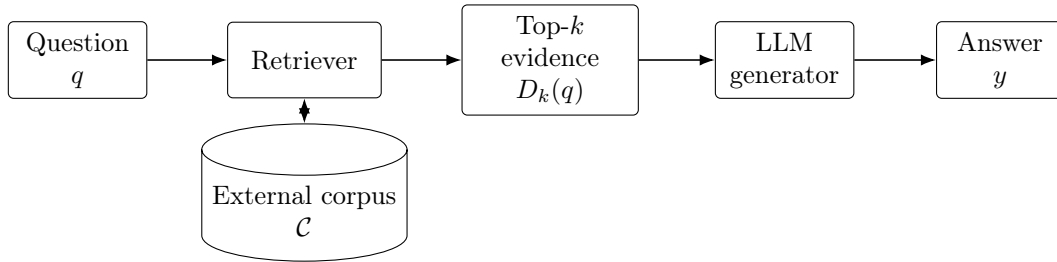


Figure 2.1: A conventional RAG pipeline. Retrieval selects a small portion of the external corpus; only those passages are available to the language model during generation.

The retrieval cutoff k controls both opportunity and cost. A larger context may contain more useful evidence, but also introduces distractors and increases the amount of text processed by the generator. This tension is mild when one passage directly answers the question and more consequential when the answer depends on several passages with different lexical and semantic relations to the original query.

2.1.2 Multi-Hop Question Answering

A single-hop question can normally be resolved through one principal factual relation or evidence source. A multi-hop question instead requires several dependent facts or operations,

$$r_1 \longrightarrow r_2 \longrightarrow \cdots \longrightarrow r_h, \quad (2.4)$$

where an intermediate result may determine what must be resolved next. The entity or value transferred between two steps is called a *bridge entity*. For example, the question *Which country is the birthplace of the author of Book X?* first requires identifying the author, then retrieving that person’s birthplace. Figure 2.2 makes this dependency explicit.

Reasoning hop count and retrieval depth describe different quantities and must not be conflated:

$$\boxed{\text{reasoning hop count} \neq \text{retrieval depth.}} \quad (2.5)$$

A four-hop benchmark question encodes four dependent reasoning operations. Retrieval depth $k = 50$, by contrast, means that up to fifty ranked passages are considered for a query. Later chapters seek to reduce the latter; they do not alter the benchmark’s compositional hop count.

2.1.3 Multi-Hop Question Answering Benchmarks

Multi-hop benchmarks operationalise compositional reasoning in different ways. Three datasets are particularly relevant to this thesis. **HotpotQA** is a Wikipedia-based benchmark designed for diverse, explainable multi-hop question answering. In addition to answers, it annotates supporting facts, enabling joint evaluation of answer prediction and evidence identification [2].

2WikiMultiHopQA constructs multi-hop questions from Wikipedia and Wikidata information and provides explicit evidence and reasoning structures. Its question types

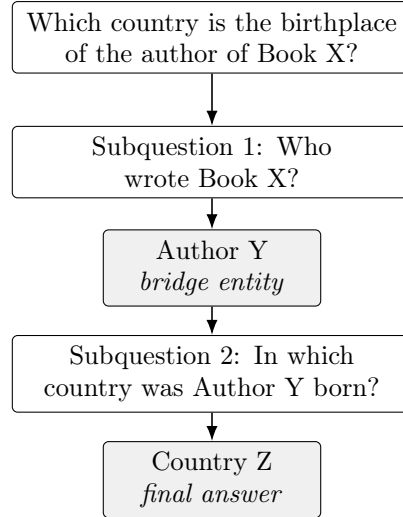


Figure 2.2: A two-hop reasoning chain. The answer to the first subquestion supplies a bridge entity needed to formulate and resolve the second information need.

include comparison and compositional chains, allowing models to be assessed at intermediate as well as final steps [3].

MuSiQue is constructed bottom-up by composing connected single-hop questions into questions requiring two to four hops. Its construction preserves dependencies among intermediate questions and supporting passages, while filtering examples that can be answered through shortcuts [4]. This structured relation between intermediate questions and evidence is also useful when studying how generated decompositions divide responsibility for retrieving the evidence.

The three benchmarks therefore provide distinct operational views of compositional difficulty. They form the empirical basis of the layer-wise probing study introduced in Chapter 3 [5], without assuming that a nominal hop label is a complete measure of all forms of difficulty.

2.1.4 Evidence and Supporting Passages

Let the gold evidence for a question be

$$E(q) = \{e_1, e_2, \dots, e_m\}, \quad (2.6)$$

and let the retriever produce the ordered ranking

$$R(q) = (d_1, d_2, \dots, d_N). \quad (2.7)$$

For $e \in E(q)$, $\text{rank}(e \mid q)$ denotes its one-based position in this ranking. Evidence retrieval is successful only to the extent that the required supporting passages appear within the part of the ranking made available downstream.

Answer correctness and evidence retrieval success are related but not equivalent. An LLM can occasionally answer correctly from parametric memory even when the annotated supporting passages are absent. Conversely, retrieving the gold passages does not guarantee that the generator will combine them correctly. For this reason, the later evaluation isolates whether required evidence is available instead of inferring retrieval quality solely from the final answer.

2.2 Retrieval for Multi-Hop Reasoning

Retrieval methods differ chiefly in how they represent queries and passages and how they compute the score $s(q, d)$. Sparse methods emphasise lexical overlap, dense methods learn semantic vector representations, and late-interaction methods retain token-level matching. These choices determine the ranking on which evidence coverage and retrieval depth are measured.

2.2.1 Sparse Retrieval

Sparse retrieval represents text using high-dimensional term vectors. In a Term Frequency–Inverse Document Frequency (TF–IDF) representation, the weight of term t in document d is

$$w(t, d) = \text{tf}(t, d) \text{idf}(t), \quad \text{idf}(t) = \log \frac{N}{\text{df}(t)}, \quad (2.8)$$

where $\text{tf}(t, d)$ is the term frequency and $\text{df}(t)$ counts documents containing t . Query and document vectors can then be compared with a cosine-style similarity. TF–IDF assigns greater influence to terms that are frequent in a document but rare throughout the corpus [6].

BM25 refines lexical scoring by saturating term-frequency gains and normalising for document length [7]:

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}}\right)}. \quad (2.9)$$

Here $f(t, d)$ is the term frequency, k_1 controls saturation, b controls document-length normalisation, and avgdl is average document length. Sparse indices are efficient and interpretable, and exact entity names often provide strong signals. Their principal limitation is lexical mismatch: a relevant passage can express the needed relation without sharing the query’s wording, especially when the original question hides an intermediate entity.

2.2.2 Dense Retrieval

Dense retrieval learns encoders that map text into a continuous vector space. Dense Passage Retrieval (DPR), for example, uses separate question and passage encoders [8]:

$$\mathbf{q} = f_q(q), \quad \mathbf{d} = f_d(d), \quad s(q, d) = \mathbf{q}^\top \mathbf{d}. \quad (2.10)$$

Passage vectors can be precomputed and searched with approximate nearest-neighbour methods. Because the representation is learned, semantically related text can match despite limited word overlap. This is valuable when a question and its evidence use different expressions.

The compact representation also creates a bottleneck. A single vector must compress all potentially relevant entities and relations in a passage. For a complex query that simultaneously implies several information needs, the dot product provides only a coarse aggregate match and may obscure which part of the query aligns with which part of a passage.

2.2.3 Late-Interaction Retrieval and Reranking

Contextualized Late Interaction over BERT (ColBERT) occupies a middle ground between lexical matching and single-vector dense retrieval [9]. It independently encodes contextual query tokens $\mathbf{q}_i$ and document tokens $\mathbf{d}_j$, then applies a maximum-similarity operation for every query token:

$$s_{\text{ColBERT}}(q, d) = \sum_i \max_j \mathbf{q}_i^\top \mathbf{d}_j. \quad (2.11)$$

This *MaxSim* interaction preserves fine-grained matches while allowing document representations to be precomputed. It is relevant to the later arm–evidence attribution problem because deciding whether a subquestion is responsible for a passage is partly a fine-grained query–passage matching task.

Retrieval systems also commonly separate inexpensive candidate generation from more expensive scoring. A fast retriever first produces a manageable candidate set; a cross-encoder or another stronger reranker jointly examines each query–passage pair and re-orders that set. Reranking can improve precision near the top but cannot recover a gold passage omitted from the candidate set. The initial retrieval stage therefore continues to constrain attainable evidence coverage.

2.2.4 Iterative and Multi-Hop Retrieval

Single-pass retrieval uses the same original question to search for every supporting passage. Iterative retrieval instead allows evidence acquired at one step to change the next query. Multi-Hop Dense Retrieval (MDR) recursively retrieves passages with a dense representation conditioned on the question and earlier evidence, enabling later searches to incorporate newly identified bridge information [10].

Baleen addresses the rapid growth of the multi-hop search space through *condensed retrieval*: after each iteration, salient information from the retrieved material is condensed into a compact context for subsequent search. It combines this mechanism with focused late interaction and a learned ordering of retrieval steps [11].

Interleaving Retrieval with Chain-of-Thought Reasoning (IRCoT) alternates the two operations. An intermediate reasoning step specifies what information is needed next; retrieved evidence then updates the context used to produce the following step [12]. These approaches differ architecturally, but share a crucial observation: retrieval can benefit from intermediate state rather than repeatedly treating the initial question as a fixed search query.

The original full question is not necessarily the best retrieval query for every evidence passage.

This observation motivates the later shift from implicit intermediate state to explicit question decomposition.

2.2.5 Evidence Coverage and Retrieval Depth

For the top- k passages $D_k(q)$, evidence coverage is the fraction of gold passages retrieved:

$$\text{Coverage @}k(q) = \frac{|E(q) \cap D_k(q)|}{|E(q)|}. \quad (2.12)$$

Complete recovery at a cutoff can be recorded as

$$\text{Complete}(q, k) = \mathbb{1}[E(q) \subseteq D_k(q)]. \quad (2.13)$$

When all required evidence is present in the ranking, the minimum depth at which it has been recovered is determined by the lowest-ranked required passage:

$$K^*(q) = \max_{e \in E(q)} \text{rank}(e \mid q). \quad (2.14)$$

Figure 2.3 illustrates these quantities. At $k = 5$, two of three gold passages have been recovered, so coverage is $2/3$. Complete coverage is first reached at $K^* = 7$.

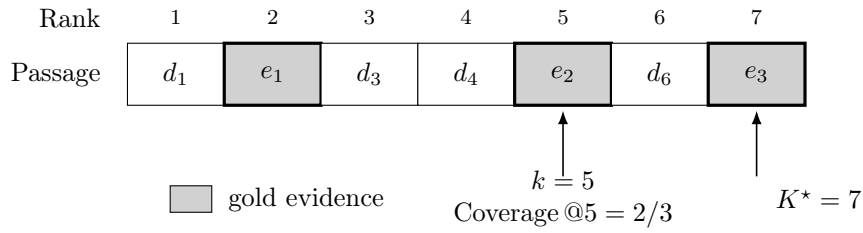


Figure 2.3: A ranked retrieval list with three gold passages. Coverage at a fixed cutoff and the minimum depth K^* capture complementary aspects of retrieval.

If at least one required passage does not occur within the evaluated search horizon, the question is *uncovered*; assigning it an artificial finite depth would falsely make failure appear efficient. K^* is thus defined for questions whose evidence is recovered, while coverage separately accounts for failures. It is a rank-based proxy for retrieval effort, not a reasoning hop count and not a direct measurement of wall-clock computation. Actual latency also depends on indexing, model architecture, batching, and hardware.

Coverage and depth must therefore be evaluated jointly. A useful retrieval strategy should expose as much of the required evidence as possible and do so near the top of its rankings:

$$\boxed{\text{high evidence coverage} \quad + \quad \text{low retrieval depth.}} \quad (2.15)$$

Here *evidence coverage* means the recovery of annotated supporting passages. It is distinct from *conformal* or *statistical coverage*, which concerns the long-run validity of prediction sets or risk guarantees in uncertainty-quantification methods.

Because the depth required for complete evidence recovery varies substantially from one question to another, applying a uniformly expensive retrieval policy is neither necessary nor efficient. This observation leads to systems that estimate what an input needs and adapt their computation accordingly.

2.3 Difficulty-Aware and Adaptive Computation

Fixed pipelines allocate the same retrieval and reasoning procedure to every input, although a direct factual question and a four-step compositional question have different needs. Adaptive systems instead obtain a signal about the input or current state and translate it into an action:

$$\boxed{\text{signal} \longrightarrow \text{adaptive computational action.}} \quad (2.16)$$

2.3.1 Query Complexity and Routing

Adaptive-RAG demonstrates complexity-aware routing for question answering. A classifier estimates whether a question is best handled without retrieval, with single-step retrieval, or with an iterative strategy, thereby avoiding the cost of a uniformly complex pipeline [13]. In generic form,

$$q \longrightarrow \widehat{d}(q) \longrightarrow \pi_{\widehat{d}}, \quad (2.17)$$

where $\widehat{d}(q)$ is an estimated difficulty or complexity level and $\pi_{\widehat{d}}$ is the selected computational policy. The important point is not a particular number of routing classes, but that a pre-answer estimate can govern retrieval, reasoning, or decomposition before their full costs are paid.

Existing routing approaches establish that question complexity is actionable. They also motivate a representational question: instead of estimating difficulty only from surface characteristics with an external classifier, could the target LLM itself already encode this information internally? This is the question taken up in Chapter 3.

2.3.2 Adaptive Retrieval During Generation

Adaptation can also occur after generation begins. Forward-Looking Active REtrieval augmented generation (FLARE) anticipates the next sentence, examines confidence in the predicted content, and triggers retrieval when low-confidence tokens indicate that more information is needed. The retrieved documents are then used to regenerate the sentence [14]. Self-RAG trains a model to produce reflection signals that control retrieval on demand and support critique of both evidence and generated text [15].

These systems illustrate a broader design principle identified in the Trust Layer survey conducted during the internship: signals become operationally useful when connected to actions such as retrieval, re-retrieval, routing, stopping, filtering, or abstention [16]. The present thesis adopts this signal-to-action perspective, but does not require a full taxonomy of uncertainty quantification. Its focus is the allocation of retrieval and decomposition effort for multi-hop questions.

2.3.3 Difficulty and Uncertainty

Difficulty and uncertainty can influence the same adaptive action, but they are not interchangeable:

$$\boxed{\text{difficulty} \neq \text{uncertainty}.} \quad (2.18)$$

Difficulty describes the amount or complexity of reasoning structurally required by a question. *Uncertainty* describes the reliability of the system’s current prediction, retrieval state, or answer. A difficult question may be answered confidently when all necessary evidence and reasoning are available; an easy question may remain uncertain because retrieval failed. The other combinations—difficult and uncertain, or easy and confident—are also possible. Chapter 3 therefore studies compositional difficulty from question representations before answer generation, rather than estimating uncertainty in an answer that has already been produced.

Estimating difficulty is only useful if the estimate changes what the system does. For a question predicted to require compositional reasoning, one possible action is to expose its latent information needs explicitly through decomposition.

2.4 Question Decomposition for Multi-Hop Reasoning

Question decomposition makes the information needs of a complex question explicit. Let

$$D(q) = \{a_1, a_2, \dots, a_m\}, \quad (2.19)$$

where each a_i is a simpler subquestion. In the remainder of this thesis, generated subquestions are also called *decomposition arms*. The notation does not imply that arms are mutually independent: some can require answers obtained by earlier arms.

2.4.1 Dependency-Aware Decomposition

In a chained decomposition, execution alternates between asking a subquestion and obtaining an intermediate answer:

$$a_1 \longrightarrow z_1 \longrightarrow a_2(z_1) \longrightarrow z_2 \longrightarrow a_3(z_2), \quad (2.20)$$

where z_i is an intermediate answer or bridge entity. Before z_i is known, a downstream arm can represent the unresolved dependency abstractly; execution resolves dependencies in order and supplies the available information. The final downstream arm obtains what is needed to answer the original question.

More generally, dependencies form a directed acyclic graph (DAG),

$$G = (V, E), \quad V = \{a_1, \dots, a_m\}, \quad (2.21)$$

whose directed edges identify which arms supply information to others. A chain is a special case of a DAG; branching allows independent information needs to be resolved in parallel and later combined. Figure 2.4 shows a generic branching example without prescribing a particular generation format or execution mechanism.

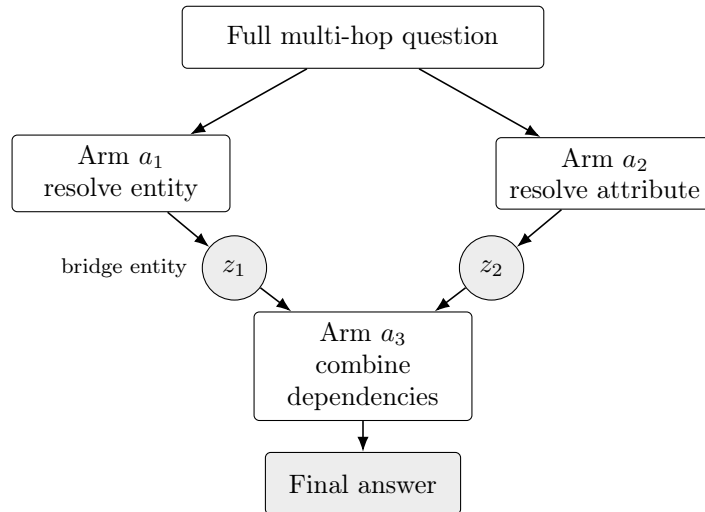


Figure 2.4: A generic dependency-aware decomposition. Arms may resolve independent needs, while directed edges carry bridge entities or intermediate answers to downstream arms. A purely sequential decomposition is a special case of this DAG.

2.4.2 The Compositionality Gap

Explicit decomposition is motivated partly by the *compositionality gap*: a model can answer the component questions needed for a problem yet fail when they are presented as one composed question. Press et al. show that this gap is not necessarily eliminated by model scaling and introduce Self-Ask, in which the model generates and answers explicit follow-up questions before producing its final response [17]. The result is important here not as a claim that decomposition always succeeds, but as evidence that exposing intermediate information needs can change a model’s ability to use knowledge it already possesses or can retrieve.

2.4.3 Earlier Question-Decomposition Approaches

Question decomposition predates modern instruction-tuned LLMs. Min et al. decompose multi-hop reading-comprehension questions into simpler single-hop questions, answer them with existing readers, and globally rescore candidate decompositions and answers. Their method reduces the amount of dedicated supervision by treating subquestion generation as span prediction [18].

Perez et al. propose an unsupervised one-to-many sequence-transduction approach that learns to produce single-hop subquestions without explicit decomposition labels. Answers from an off-the-shelf question-answering model are subsequently recomposed into the final answer [19]. Together, these methods establish decomposition as an intermediate representation that can connect a complex question to reusable single-hop components.

2.4.4 LLM-Based Decomposition

The prompting capabilities of LLMs enable decomposition without a fixed task-specific parser. Successive Prompting repeatedly generates a simpler question, answers it, and uses the result to continue until the complex problem is resolved [20]. Least-to-Most Prompting first decomposes a problem and then solves the resulting subproblems sequentially, allowing earlier solutions to support later ones [21]. Decomposed Prompting treats the solution as a modular program in which a controller delegates subproblems to specialised prompts or modules [22].

Their common pattern is

complex question $\longrightarrow$ explicit simpler subproblems $\longrightarrow$ dependency-aware
solution $\longrightarrow$ final aggregation.

Because an LLM’s decomposition is generated rather than fixed, it is itself a configurable reasoning component. Different instructions or examples can expose different subproblems, and decomposition quality consequently affects all later retrieval and reasoning stages.

2.4.5 From Answer-Oriented to Retrieval-Oriented Decomposition

Most decomposition methods are primarily evaluated through downstream task or answer performance. The perspective adopted in this thesis adds a narrower question: *does*

the decomposition make the required evidence easier to retrieve? Suppose a supporting passage e appears deep in the full-question ranking but near the top for a targeted arm:

$$\text{rank}(e \mid q) = 80, \quad \text{rank}(e \mid a_i) = 4. \quad (2.22)$$

The underlying reasoning dependency has not disappeared. Rather, the arm states a narrower information need and can remove lexical or semantic interference from unrelated parts of the full question.

This perspective does not imply that previous work ignores retrieval. Self-Ask can call a search engine for follow-up questions, and IRCOT explicitly couples intermediate reasoning with search [17, 12]. The contribution developed later is more specific: it makes evidence coverage and maximum retrieval depth explicit criteria for evaluating and optimising a generated decomposition.

2.5 Adaptive Decomposition and Efficient Evidence Retrieval

If decomposition is to improve retrieval, it must expose each evidence need without discarding difficult passages. This requires more than retrieving separately for every arm: the evaluation must know which arm is expected to retrieve which piece of evidence and must aggregate those arm-level results into a valid question-level measure.

2.5.1 The Evidence-Responsibility Problem

Consider a decomposition $D(q) = \{a_1, \dots, a_m\}$. Its gold evidence is $E(q) = \{e_1, \dots, e_n\}$. When retrieving with the full question, all evidence is associated with a single query. Under decomposition, evaluating every arm against every passage would penalise a focused arm for evidence unrelated to its subproblem. Let the responsibility relation be

$$z_{ij} = \begin{cases} 1, & \text{if arm } a_i \text{ is responsible for evidence } e_j, \\ 0, & \text{otherwise.} \end{cases} \quad (2.23)$$

The evidence assigned to arm a_i is then

$$E_i = \{e_j : z_{ij} = 1\}. \quad (2.24)$$

For an arm whose assigned evidence is retrieved, its required depth is

$$K_i = \max_{e \in E_i} \text{rank}(e \mid a_i), \quad (2.25)$$

and the decomposition’s maximum depth, taken over arms with non-empty assigned evidence, is

$$K_{\text{decomp}} = \max_i K_i. \quad (2.26)$$

As with K^* , uncovered assignments must be recorded as coverage failures rather than silently given a favourable finite depth.

The central difficulty is determining z_{ij} for arbitrary generated decompositions, whose wording and number of arms are not fixed in advance. MuSiQue is useful for developing this evaluation because its constructed question graph relates intermediate questions to supporting evidence [4]. Chapter 4 develops the evidence-aware evaluator and arm-evidence attribution mechanism; the present chapter deliberately leaves its implementation unspecified.

2.5.2 Coverage–Depth Trade-Off

A small depth is meaningful only when the associated evidence has actually been recovered. Consider a method A with high coverage and maximum depth 10, and a method B with lower coverage and maximum depth 3. Method B is not automatically better: its shallow depth may be an artefact of missing the passages that are hardest to retrieve. The objectives are therefore simultaneous,

$$\max \text{ Coverage}, \quad \min K_{\max}. \quad (2.27)$$

At equal coverage, lower depth is preferable; at equal depth, higher coverage is preferable.

Figure 2.5 presents this as a schematic Pareto problem. The points are illustrative methods, not experimental measurements. A method is unambiguously improved when it moves upward without moving right, or leftward without moving down. Otherwise, the comparison exposes an explicit trade-off rather than collapsing failures and efficiency into one number.

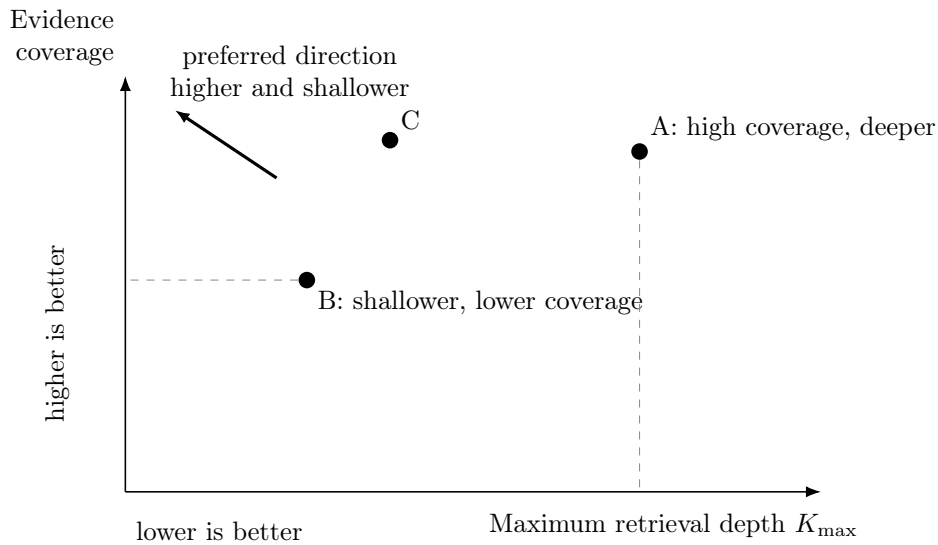


Figure 2.5: Conceptual comparison of evidence coverage and maximum retrieval depth. Movement toward the upper-left is preferred. Labels A–C are schematic and do not represent experimental results.

2.5.3 Evaluating the Quality of a Decomposition

A retrieval-oriented evaluator for an arbitrary decomposition must answer five questions. First, does the set of arms preserve the evidence required by the original question? Second, is each passage assigned to an appropriate arm? Third, does that arm retrieve its evidence at a shallower rank? Fourth, can the arm-level results be aggregated without hiding uncovered evidence? Finally, does the resulting question-level metric provide a useful signal for improving the decomposer?

Existing work demonstrates that decomposition can improve reasoning and that intermediate queries can guide retrieval. The perspective adopted in this thesis is to make evidence coverage and maximum retrieval depth explicit evaluation and optimisation criteria for the decomposition. Chapter 4 develops the evaluator and arm–evidence attribution, while Chapter 5 uses this evaluation as feedback for prompt optimisation.

2.6 Research Gap and Thesis Positioning

The preceding literature connects complex questions to adaptive computation and decomposition, but leaves two questions that structure the experimental chapters of this thesis.

2.6.1 Anticipating Compositional Difficulty

Adaptive systems use query complexity or runtime confidence to select different policies. Many signals come from external classifiers, surface properties, or states observed only after generation has begun. This thesis instead asks whether the internal representation formed by the target LLM already contains a usable pre-answer signal about the question’s compositional demands.

Does the LLM already encode how much compositional reasoning a question requires?

Chapter 3 addresses this question through layer-wise probing of question representations. It separates compositional difficulty from expected answer correctness and evaluates the former without requiring an answer to be generated first.

2.6.2 Optimizing Decomposition for Retrieval Efficiency

Question decomposition exposes intermediate information needs, and iterative retrieval demonstrates that such intermediate state can improve search. The remaining issue is whether an arbitrary generated decomposition can be judged and improved specifically as a set of retrieval queries.

Can a decomposition preserve evidence coverage while reducing the maximum depth required to retrieve that evidence?

Answering this requires three capabilities: determining which evidence each arm is responsible for, evaluating decompositions jointly through coverage and depth, and adapting the decomposition-generation strategy according to those metrics. Chapter 4 develops the evidence-aware evaluation and attribution required by the first two. Chapter 5 then uses the resulting signal to optimise the decomposition prompt.

Conclusion

This chapter presented the background necessary for the experimental work in the remainder of the thesis. Multi-hop question answering requires both reasoning across several pieces of evidence and retrieval mechanisms capable of exposing that evidence efficiently. Existing work approaches these demands through stronger retrievers, iterative search, adaptive computation, and question decomposition.

Two questions remain central. The first is whether multi-hop difficulty can be identified directly from the model before answering, which is investigated in Chapter 3. The second is whether decomposition can be evaluated and optimised to preserve evidence coverage while reducing retrieval depth, which motivates Chapters 4 and 5.

Chapter 3

A Systematic Survey of the Trust Layer of RAG

Introduction

Conclusion

Chapter 4

chapter 4

Introduction

Conclusion

Chapter 5

chapter 5

Introduction

Conclusion

Conclusion and outlook

This internship project aimed to design

Bibliography

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.
- [2] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning, “HotpotQA: A dataset for diverse, explainable multi-hop question answering,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, 2018.
- [3] X. Ho, A.-K. D. Nguyen, S. Sugawara, and A. Aizawa, “Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps,” in *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020.
- [4] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, “MuSiQue: Multihop questions via single-hop question composition,” *Transactions of the Association for Computational Linguistics (TACL)*, vol. 10, pp. 539–554, 2022.
- [5] O. Abdelhedi, O. R. Heidari, S. Chaari, X. Chen Zhu, M. Hamed, D. Azzam, and Y. Yaakoubi, “Probing compositional question difficulty in language model representations,” in *IEEE AIDIST 2026 Conference*, 2026. Accepted, to appear.
- [6] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [7] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [8] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, “Dense passage retrieval for open-domain question answering,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6769–6781, 2020.
- [9] O. Khattab and M. Zaharia, “ColBERT: Efficient and effective passage search via contextualized late interaction over BERT,” in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 39–48, 2020.
- [10] W. Xiong, X. L. Li, S. Iyer, J. Du, P. Lewis, W. Y. Wang, Y. Mehdad, W.-t. Yih, S. Riedel, D. Kiela, and B. Oğuz, “Answering complex open-domain questions with

- multi-hop dense retrieval,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [11] O. Khattab, C. Potts, and M. Zaharia, “Baleen: Robust multi-hop reasoning at scale via condensed retrieval,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.
- [12] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, “Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 10014–10037, 2023.
- [13] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, “Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 7036–7050, 2024.
- [14] Z. Jiang, F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, and G. Neubig, “Active retrieval augmented generation,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7969–7992, 2023.
- [15] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-RAG: Learning to retrieve, generate, and critique through self-reflection,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [16] O. R. Heidari, S. Chaari, O. Abdelhedi, D. Azzam, M. Hamed, X. Chen Zhu, and Y. Yaakoubi, “The trust layer of retrieval-augmented generation: A systematic survey of uncertainty quantification, verification, and guarantees,” in *IEEE AIDIST 2026 Conference*, 2026. Accepted, to appear.
- [17] O. Press, M. Zhang, S. Min, L. Schmidt, N. A. Smith, and M. Lewis, “Measuring and narrowing the compositionality gap in language models,” in *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 5687–5711, 2023.
- [18] S. Min, V. Zhong, L. Zettlemoyer, and H. Hajishirzi, “Multi-hop reading comprehension through question decomposition and rescoring,” in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 6097–6109, 2019.
- [19] E. Perez, P. Lewis, W.-t. Yih, K. Cho, and D. Kiela, “Unsupervised question decomposition for question answering,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 8864–8880, 2020.
- [20] D. Dua, S. Gupta, S. Singh, and M. Gardner, “Successive prompting for decomposing complex questions,” in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 1251–1265, 2022.
- [21] D. Zhou, N. Schärli, L. Hou, J. Wei, N. Scales, X. Wang, D. Schuurmans, C. Cui, O. Bousquet, Q. V. Le, and E. H. Chi, “Least-to-most prompting enables complex reasoning in large language models,” in *International Conference on Learning Representations (ICLR)*, 2023.

-
- [22] T. Khot, H. Trivedi, M. Finlayson, Y. Fu, K. Richardson, P. Clark, and A. Sabharwal, “Decomposed prompting: A modular approach for solving complex tasks,” in *International Conference on Learning Representations (ICLR)*, 2023.

Netography

- [N1] Concordia University, “The birth of Concordia: Loyola College and Sir George Williams University merged on 24 August 1974.” <https://www.concordia.ca/cunews/main/stories/2015/08/24/archives-loyola-and-sir-george-williams-merge.html>. Accessed: 2026-08-13.
- [N2] Concordia University, “Sir George Williams University history.” <https://www.concordia.ca/offices/archives/stories/sgw.html>. Accessed: 2026-08-13.
- [N3] Concordia University, “Concordia fast facts: student enrolment 2024–25.” <https://www.concordia.ca/about/fast-facts.html>. Accessed: 2026-08-13.
- [N4] Concordia University, “First engineering faculty in Canada named after a woman: business leader Gina Cody makes historic \$15 million donation.” <https://www.concordia.ca/news/stories/2018/09/24/gina-cody-makes-historic-eponymous-15m-donation-to-have-canadas-first-engineering-faculty-named-after-a-woman.html>. Published 24 September 2018. Accessed: 2026-08-13.
- [N5] Concordia University, “Gina Cody — notable Concordians.” <https://www.concordia.ca/profiles/offices/advancement/notable-concordians/gina-cody.html>. Accessed: 2026-08-13.
- [N6] Concordia University, “About the school — gina cody school of engineering and computer science.” <https://www.concordia.ca/ginacody/about.html>. Accessed: 2026-08-13.